Oxford College of Emory University

# Chapter 14, Part 1

## Money, Liquidity, and Banking

Guilherme Paiva Pinto

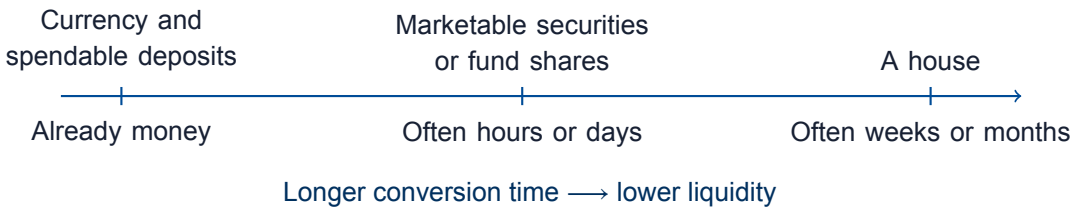
October 27, 2026

# Roadmap

1. Money, liquidity, and the money supply
2. Banks, reserves, and T-accounts
3. Bank runs and safeguards
4. Moral hazard and shadow banking

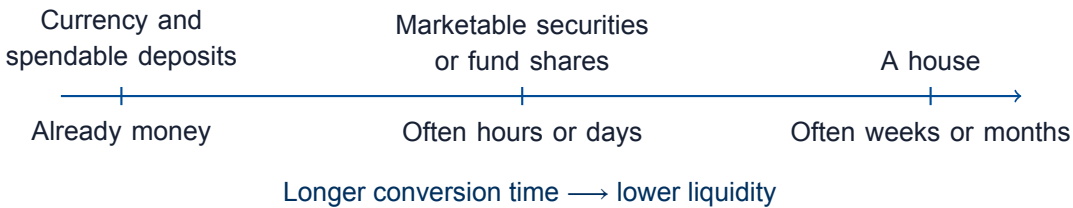
# Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.



# Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.
- **Liquidity**: how quickly something can become spendable money, with little cost or loss of value.



# The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.

**M2 = M1 + additional near-money.** Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

# The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.

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- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.
- **M2, the broader measure**: M1 plus small time deposits and retail money market mutual fund balances.

**M2 = M1 + additional near-money.** Near-money is readily convertible into spendable balances.

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# The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
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  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.
- **Store of value**: a way to carry purchasing power into the future.
  - Save today's earnings for next month's groceries. Inflation can erode their value.

# Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

**Commodity money**

Pay with a portion of the crop accepted as money. It also has value as food.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

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Pay with a warehouse receipt redeemable for a specified quantity of grain.

**Fiat money**
Pay with officially accepted money that promises no commodity redemption.

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# What Banks Do

- Banks **accept deposits** and provide payment services.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

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- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.

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# What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.
- Deposits can be withdrawn quickly; many loans and investments repay over years.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

# Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

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- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.

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# Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.
- **Reserve ratio**: bank reserves divided by the bank's deposits.

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# T-Accounts

A **T-account** shows **what a bank has on the left** and **what it owes on the right**.

| What the bank has | | What the bank owes | |
|---|---|---|---|
| Loans | $1,200,000 | Deposits | $1,000,000 |
| Reserves | $100,000 | | |

Total on the left: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' money, which absorbs losses first.

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

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**Reserve ratio** $= \frac{100{,}000}{1{,}000{,}000} = 10\%.$

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# Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

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What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

# Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.

Source: Federal Reserve Board, SVB review, April 2023.

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- The bank has lent or invested much of that money. It cannot get it all back immediately.

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# Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.
- The bank has lent or invested much of that money. It cannot get it all back immediately.
- Selling investments quickly can mean losses. This can make more customers want to leave.

Source: Federal Reserve Board, SVB review, April 2023.

# Silicon Valley Bank: March 2023

- **2022**: higher interest rates made SVB's older, long-term bonds worth less.

"Silicon Valley Bank shut by California regulator"
Reuters, March 10, 2023.

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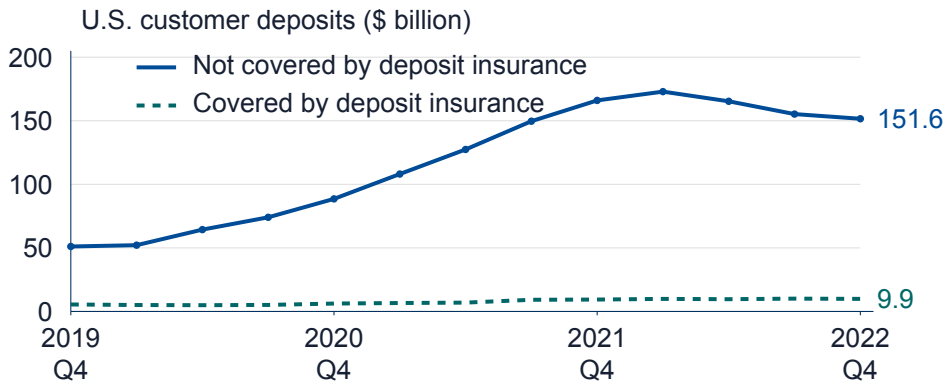
# Silicon Valley Bank: March 2023

- **2022**: higher interest rates made SVB's older, long-term bonds worth less.
- **March 8–9**: SVB announced losses. Worried customers withdrew more than $40 billion the next day.
- **March 10**: the bank could not meet the rush of withdrawals. Regulators closed it.

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# Silicon Valley Bank: Customer Deposits




Original graph. Data: Federal Reserve Board, April 2023 SVB review, Figure 7 data; Call Report.

# Bank Regulation

- **Deposit insurance**: the FDIC, created in 1933, protects eligible deposits up to $250,000 per depositor, insured bank, and ownership category.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

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- **Interest on reserve balances (IORB)**: the Fed pays interest on money banks keep in their accounts at the Fed.
- **Capital requirements**: owners must put in their own money to absorb losses first.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

# Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.

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- Customers may watch risks less closely if their deposits are insured. Owners may gain from risky bets while others bear some losses.
- **Capital requirements** make owners risk more of their own money; bank oversight helps enforce the rules.

Source: Federal Reserve Board, regulatory capital.

# Shadow Banking

- **Shadow banking**: lending and borrowing through firms that are not ordinary deposit banks.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

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## The 2008 crisis

- September 15: **Lehman Brothers** went bankrupt.
- September 16: losses at **Reserve Primary Fund** led investors to demand their money back.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

# Chapter 14, Part 2

Oxford College of Emory University

# Money Creation and Central Banks

Guilherme Paiva Pinto

October 29, 2026

# Roadmap

1. Explain how banks create money.
2. Calculate and interpret the simple money multiplier.
3. Understand how central banks work and identify their policy tools.
4. Understand how lessons from financial crises shaped monetary policymaking.

Chapter 14, Part 2 · Section 1

# Explain how banks create money.

Trace deposits, loans and payments, distinguishing changes in the money supply from changes in the monetary base.

# Determining the Money Supply

## Money supply = public currency + checkable deposits

Simplified measure: M = C + D

| The same $1,000 | Currency (C) | Deposits (D) | Money (M) |
|---|---|---|---|
| Before depositing | $1,000 | $0 | $1,000 |
| After depositing | | | |

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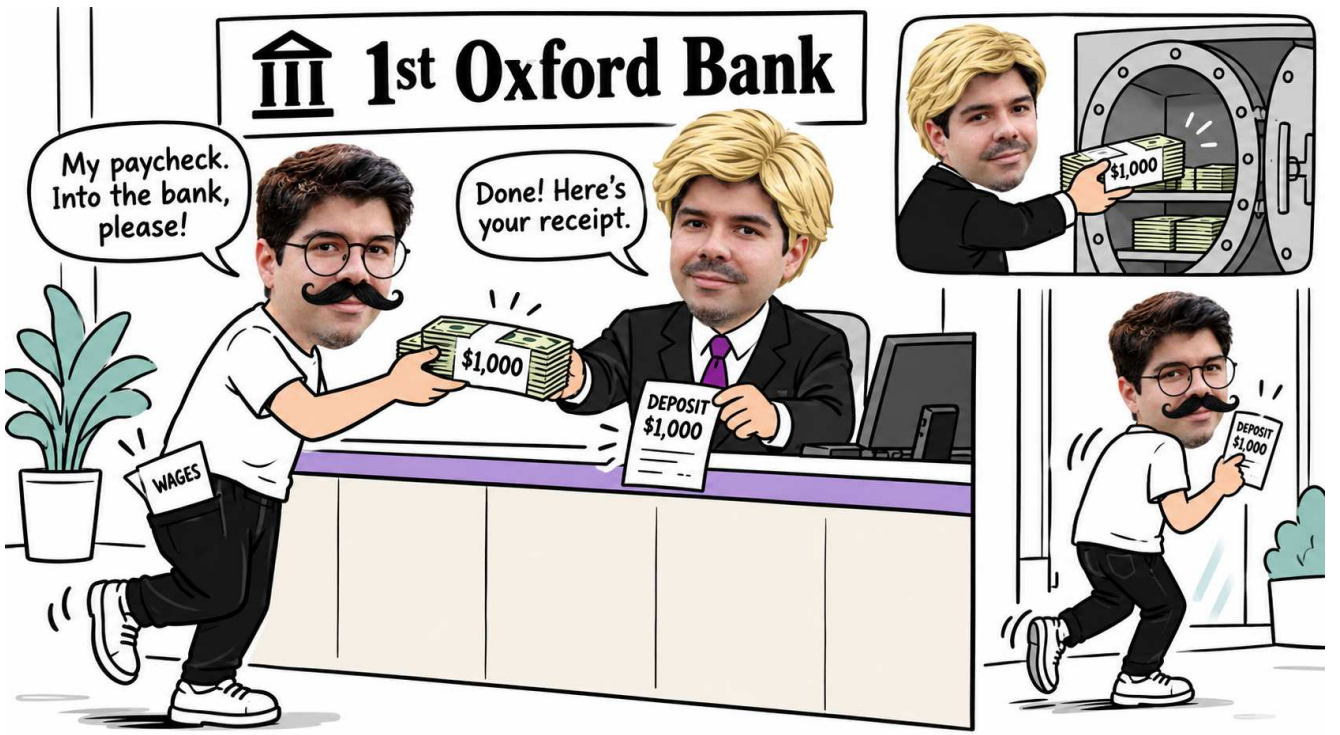
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The deposit changes the form of money, not its total.

# How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%




The cash deposit changes the form of $1,000, not the money total.

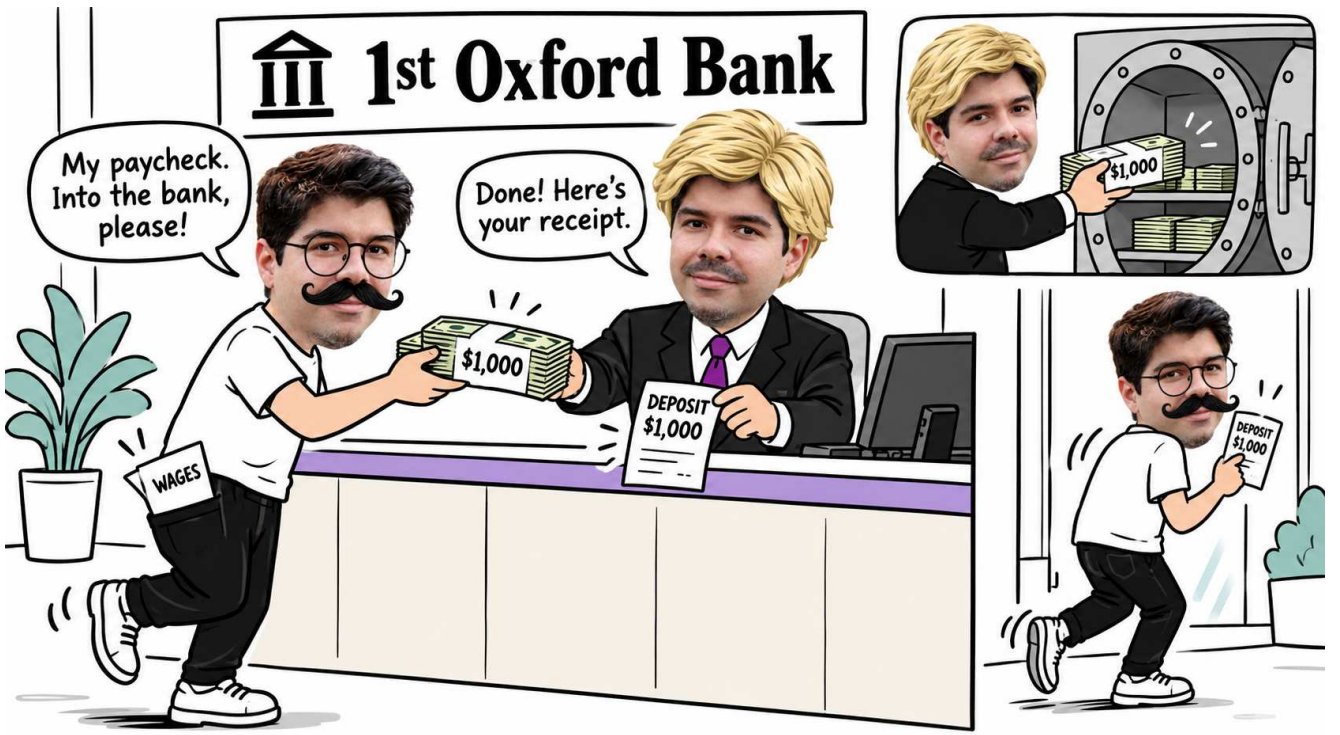
## How banks create money

| Stage | Currency in circulation | Checkable bank deposits | Money supply |
|---|---|---|---|
| **First stage**<br>Cash held by the public | $1,000 | $0 | $1,000 |
| **Second stage**<br>First deposit → $900 loan and payment | | | |
| **Third stage**<br>$900 redeposit → $810 new loan | | | |

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

# How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%




First bank: $100 required reserves; $900 available to lend.

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The first bank lends $900 in cash.

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The borrower pays the same $900 to the supplier.

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| **Third stage**<br>$900 redeposit → $810 new loan | $810 | $1,900 | $2,710 |

The next bank receives $900, keeps $90 and lends $810.

The third stage includes the **next $810 loan**.

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

Chapter 14, Part 2 · Section 2

# Calculate and interpret the simple money multiplier.

Determine total deposit expansion and explain the assumptions that limit the model’s application to actual banking.

# Reserves and Repeated Deposit Creation

**Excess reserves** = reserves − required reserves.

rr = 10% · Each row follows lending and payment to the next bank.

| Bank | Deposits | Reserves kept | New loan |
|---|---|---|---|
| A | $1,000 | | |
| B | | | |
| C | | | |
| … | … | … | … |

# Reserves and Repeated Deposit Creation

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At Bank A: $1,000 − (0.10 × $1,000) = **$900**.

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| B | $900 | $90 | $810 |
| C | $810 | $81 | $729 |
| … | … | … | … |

# The Simple Money Multiplier

Initial cash deposit: \$1,000 · rr = 0.10

$$D = 1{,}000 + 1{,}000(1 - rr) + 1{,}000(1 - rr)^2 + \ldots$$

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**$10,000 total deposits** includes the original $1,000.

**$9,000 additional deposits** are created through lending.

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Under these assumptions, \$1 of reserves supports \$10 of deposits.

# Why Reality Is More Complicated

**Full redeposit.** People deposit the payments they receive.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

# Why Reality Is More Complicated

**Full redeposit.** People deposit the payments they receive.

**Only required reserves are retained.** Banks keep rr × deposits and lend the remaining amount.

At rr = 10%: a $900 deposit → $90 reserves + $810 loan.

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# Why Reality Is More Complicated

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At rr = 10%: a $900 deposit → $90 reserves + $810 loan.

More cash held by the public or more excess reserves
means less deposit expansion.

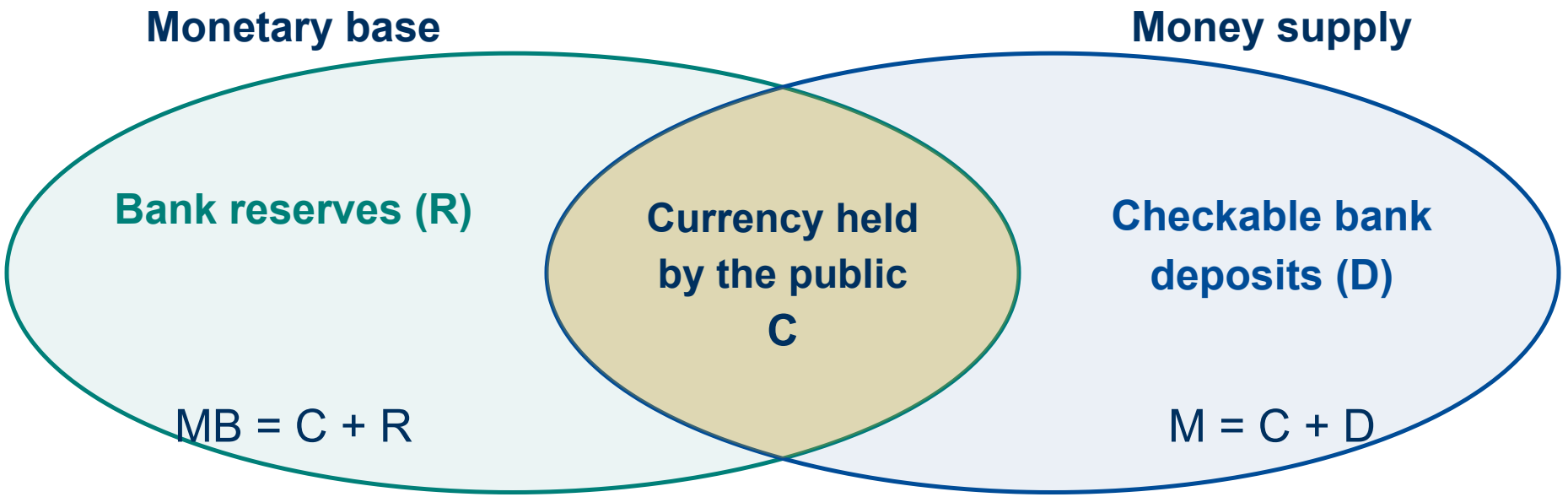
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**1/rr** is the model's simple deposit multiplier.

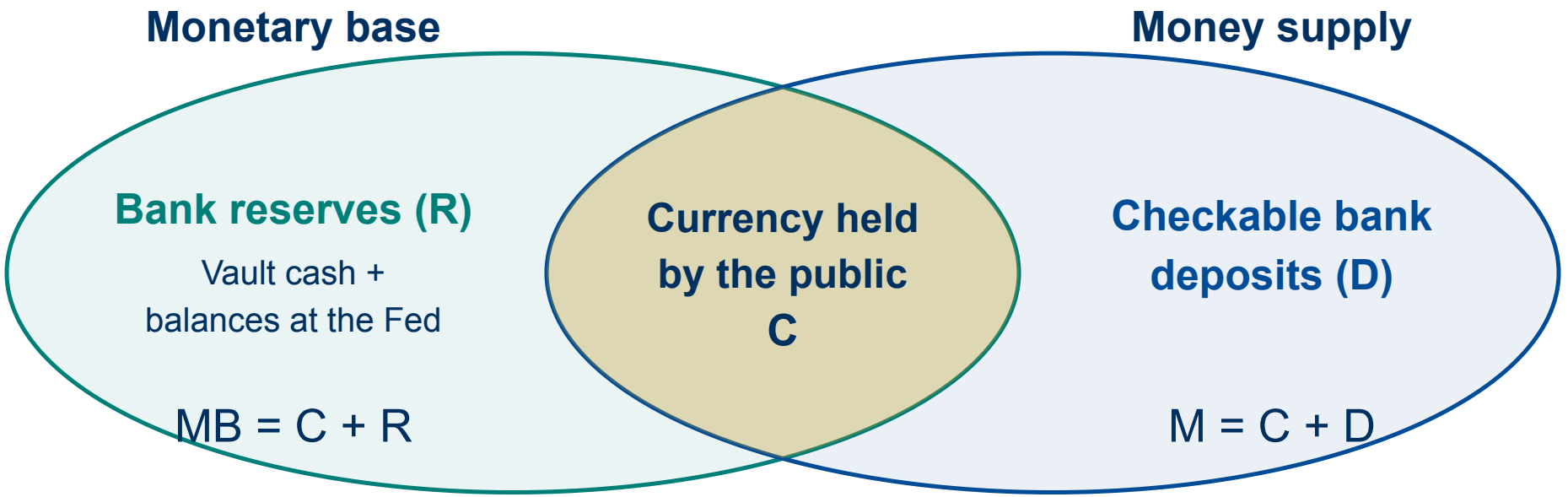
**rr = 10%** is our example; the U.S. statutory requirement is 0%.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

# Monetary Base and Money Supply

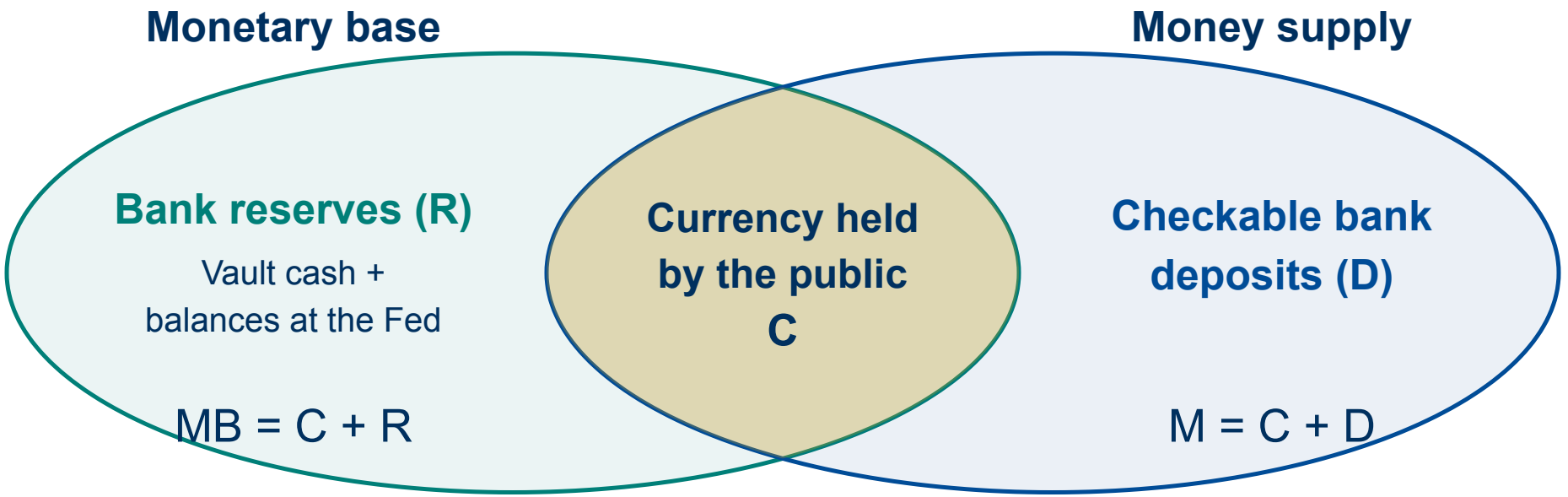


# Monetary Base and Money Supply




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$$\text{Money multiplier} = \frac{M}{MB}$$

Next: how Fed operations change reserves and the monetary base.

# Money Creation Checkpoint

$1,000 of existing public cash · rr = 10% · Full lending and redeposit

### 1. Cash becomes a deposit.

Does the money supply rise immediately?

### 2. Lending repeats across banks.

What are total deposits and the net increase in money?

### 3. Banks make loans.

Does lending itself create monetary base?

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Deposits: $10,000. Net increase in M: $9,000.

### 3. Banks make loans.

Does lending itself create monetary base?

No. It creates deposits; it does not create base money.

Chapter 14, Part 2 · Section 3

# Understand how central banks work and identify their policy tools.

Explain the roles of the discount rate, interest on reserve balances and open-market operations.

# The Federal Reserve

**The Federal Reserve (the Fed)** is the U.S. central bank.

**Its monetary-policy goals:** maximum employment and stable prices.

Federal Reserve, System Structure

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**Its monetary-policy goals:** maximum employment and stable prices.

**Monetary policy:** influence borrowing and spending to help reach those goals.

**The tools we will study:**

- Loans from the Fed to banks
- Interest paid on banks' balances at the Fed
- Purchases and sales of bonds

Federal Reserve, System Structure

# The Federal Reserve System




1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

12 Reserve Banks. San Francisco also serves Alaska and Hawaii.

## Board of Governors

**7 governors** · Washington, D.C.

Federal Reserve, System Structure · Federal Reserve, Governor Selection · Map: Natural Earth

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## Board of Governors

**7 governors** · Washington, D.C.

**Each governor:**
The U.S. President nominates;
the U.S. Senate confirms.

**Full term: 14 years.**
One term ends every two years.

**Chair:** chosen from the governors by the same process, for four years.

Federal Reserve, System Structure · Federal Reserve, Governor Selection · Map: Natural Earth

# The Federal Reserve System




1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

12 Reserve Banks. San Francisco also serves Alaska and Hawaii.

## Board of Governors

**7 governors** · Washington, D.C.

**Each governor:**
The U.S. President nominates;
the U.S. Senate confirms.

**Full term: 14 years.**
One term ends every two years.

**Chair:** chosen from the governors by the same process, for four years.

**Why regional banks?** Share responsibility and bring local economic conditions into national decisions.

Federal Reserve, System Structure · Federal Reserve, Governor Selection · Map: Natural Earth

# The Federal Open Market Committee

FOMC = the Fed’s monetary-policy committee.

**7**
Board governor seats
Permanent votes

+

**1**
New York Fed president
Permanent vote

+

**4**
Other regional presidents
One-year voting turns

Federal Reserve, FOMC

# The Federal Open Market Committee

FOMC = the Fed’s monetary-policy committee.

| 7 | + | 1 | + | 4 |
|---|---|---|---|---|
| Board governor seats | | New York Fed president | | Other regional presidents |
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**Chooses whether to raise, lower or hold the federal funds rate target range.**

Directs bond purchases and changes in the Fed’s bond holdings.

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**For our activity:** the statement explains the decision and goals; the implementation note gives the operating steps.

The Board of Governors sets reserve interest and approves discount rates.

Federal Reserve, FOMC

# Regional Presidents in the FOMC

One president from each group votes for one year.

| Reserve Bank group | 2026 voting president |
|---|---|
| Boston, Philadelphia, Richmond | **Anna Paulson** · Philadelphia |
| Cleveland, Chicago | **Beth M. Hammack** · Cleveland |
| Atlanta, St. Louis, Dallas | **Lorie K. Logan** · Dallas |
| Minneapolis, Kansas City, San Francisco | **Neel Kashkari** · Minneapolis |

Federal Reserve, FOMC · St. Louis Fed, Inside the FOMC

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**The New York Fed president always votes.** The seat passes to each successor. Its trading desk carries out FOMC market operations.

All 12 regional presidents discuss policy; seven participate without a vote that year.

Federal Reserve, FOMC · St. Louis Fed, Inside the FOMC

# What the Fed Does

Monetary policy · financial stability and supervision · payments

## The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

New York Fed, Monetary Policy Implementation

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**FFR:** the federal funds rate—the interest rate in this market.

New York Fed, Monetary Policy Implementation

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Monetary policy · financial stability and supervision · payments

## The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

**FFR:** the federal funds rate—the interest rate in this market.

| Policy tool | What changes |
|---|---|
| Discount rate | Cost of borrowing from the Fed |
| Interest on reserve balances | Return on eligible balances at the Fed |
| Open-market operations | Securities and bank reserve balances |

Next week: interest rates, spending, and aggregate demand.

New York Fed, Monetary Policy Implementation

# The Discount Window and Reserve Interest

## Discount window

The Fed lends to eligible banks against collateral.

## Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

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Cheaper Fed funding; borrowing becomes more attractive.

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Costlier Fed funding; borrowing becomes less attractive.

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**A reservation rate:**
a safe alternative return to lending.

Higher IORB raises that benchmark; lower IORB reduces it.

IORB influences lending rates; it is not a fixed minimum rate on every loan.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

# Open-Market Purchases

The Fed buys $1,000 of government bonds from a commercial bank.




Purchase: reserves enter the bank.

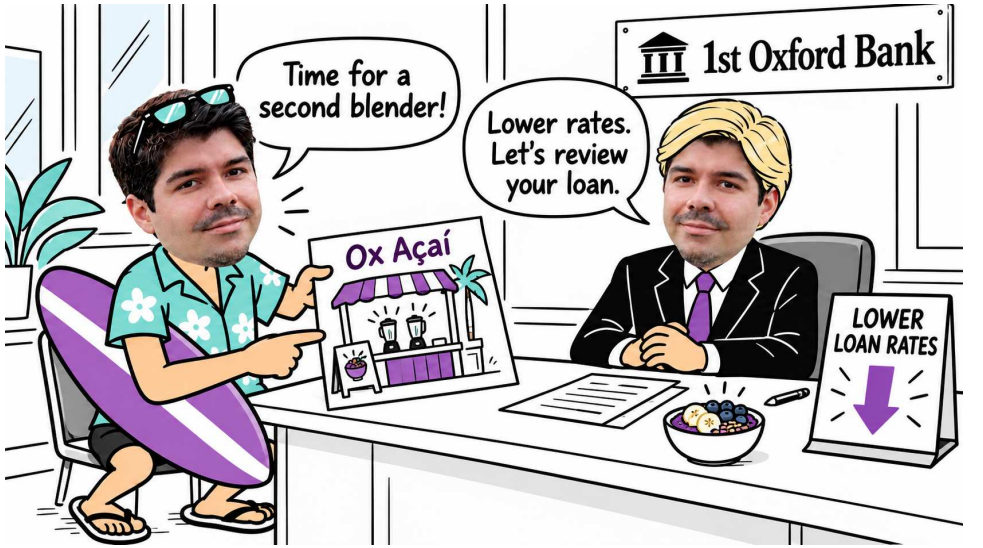
New York Fed, Monetary Policy Implementation

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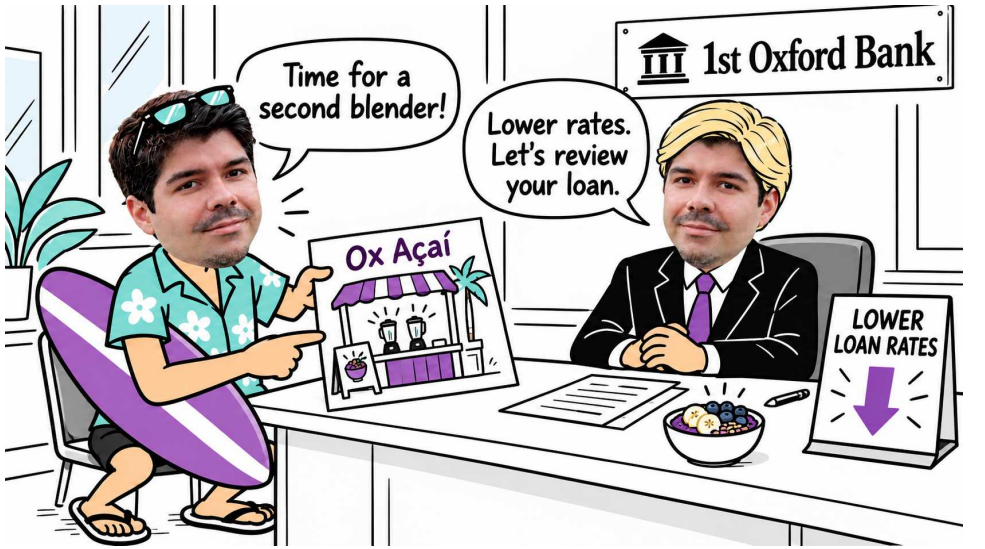
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New York Fed, Monetary Policy Implementation

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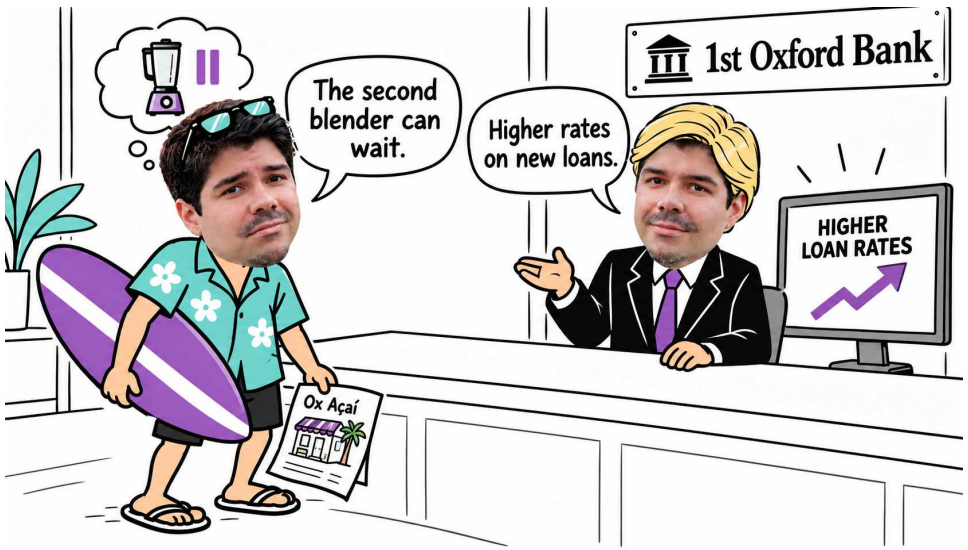
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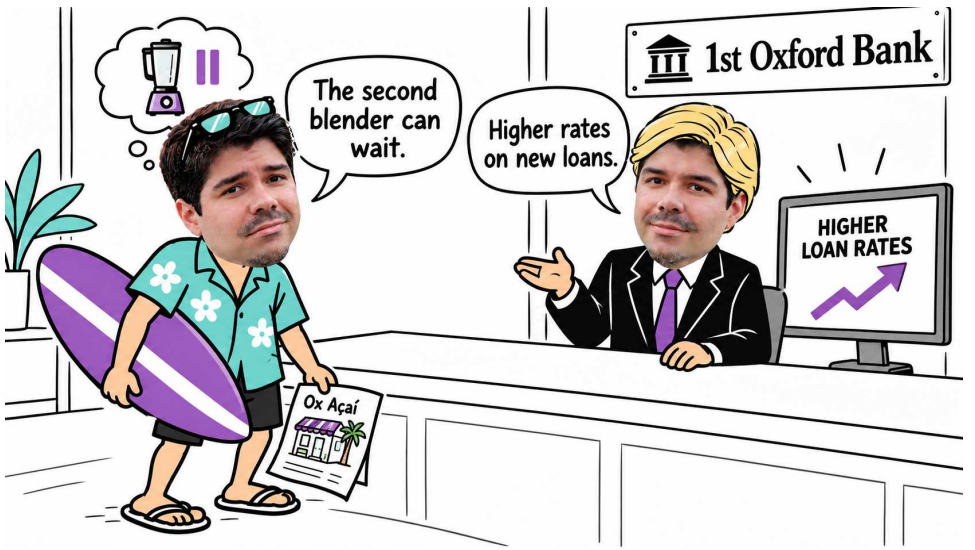
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New York Fed, Monetary Policy Implementation

# The ECB and the Federal Reserve

ECB = European Central Bank · Eurosystem = ECB + euro-area national central banks.

| | Federal Reserve | ECB / Eurosystem |
|---|---|---|
| Central board | 7 Board governor seats | 6 Executive Board members |
| Policy committee | | |
| Operating network | | |
| Main policy goals | | |

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

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| Operating network | 12 regional Reserve Banks | Euro-area national central banks |
| Main policy goals | Maximum employment; stable prices | Price stability is primary |

Both use policy rates, lending facilities, and asset operations.

The ECB’s deposit facility offers a comparable remuneration benchmark.

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

Chapter 14, Part 2 · Section 4

# Understand how lessons from financial crises shaped monetary policymaking.

Connect historical crises to changes in central banks’ responsibilities and policy approaches.

# Bank Runs and the Creation of the Fed

## 1907–1913: panic and the Fed's creation

A failed bet on copper-company shares helped trigger panic in 1907. People rushed to withdraw deposits, and banks struggled to find cash. The Fed was created on **December 23, 1913** to help meet these cash needs.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

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## 1930–1934: Great Depression bank runs

New runs helped force thousands of banks to close. Lending collapsed, and the Fed failed to stop the crisis. Federal deposit insurance was created in **1933** and began in **1934**.

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**Policy lesson:** The Fed provides emergency bank loans; the FDIC protects eligible deposits.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

# Stagflation and High Interest Rates

## 1970s: stagflation

High inflation came with weak growth and high unemployment. Oil-price shocks made both problems worse.

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Under Paul Volcker, short-term interest rates rose to nearly **20%**. Borrowing became much more expensive, slowing spending.

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Under Paul Volcker, short-term interest rates rose to nearly **20%**. Borrowing became much more expensive, slowing spending.

## 1981–1983: costs and results

A deep recession pushed unemployment to nearly **11%** in 1982. Annual inflation fell from **13.5% in 1980** to **3.2% in 1983**.

**Policy lesson:** Lower inflation came with weaker spending and serious job losses.

Fed History: Great Inflation · Fed History: Recession of 1981–82 · BLS: Annual CPI Inflation

# The 2008 Crisis and the Fed’s Response

**The problem:** Investment firms and funds relied on short-term borrowing. Home-loan losses made their lenders pull their money out—a run outside ordinary banks.

### 2007–2008: discount window and emergency loans

Cheaper, longer loans for banks. Separate emergency programs also lent to some investment firms.

Federal Reserve: Emergency Lending, March 2008 · Fed History: Interest on Reserves · Federal Reserve: Bond Purchases, November 2008

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### 2008–2009: larger open-market purchases

Bought large amounts of longer-term bonds, including bonds backed by home loans, to lower borrowing costs and support spending.

**Policy lesson:** The crisis brought broader Fed lending, larger bond purchases and a new tool for guiding rates.

Federal Reserve: Emergency Lending, March 2008 · Fed History: Interest on Reserves · Federal Reserve: Bond Purchases, November 2008

# Reference Definitions and Examples

Money, income, wealth, and bank accounting.

# Money, Income, and Net Wealth

| Concept | Meaning | Example |
|---|---|---|
| Money | Readily spendable assets | $400 in currency and deposits |
| Income | Receipts over a period | $900 earned this month |
| Net wealth | Assets minus liabilities | $5,000 − $2,000 = $3,000 |

# A Smoothie Company Balance Sheet

| Assets | Liabilities and equity |
| --- | --- |
| Cash and inventory | Loans and other obligations |
| Equipment; a building if owned | Equity: the owners' residual claim |

**Assets = liabilities + equity.**

# Liquidity and Solvency

| Issue | Question | Possible problem |
|---|---|---|
| Liquidity | Can the bank pay now? | Assets cannot all be sold quickly at little cost. |
| Solvency | Do assets cover liabilities? | Losses can exhaust equity. |

Forced sales can turn liquidity pressure into losses.

# A Cash Deposit

| Assets ($) | Liabilities and equity ($) |
|---|---|
| Reserves: 200 | Deposits: 1,000 |
| Loans: 900 | Equity: 100 |
| Total: 1,100 | Total: 1,100 |

A customer deposits $200 of currency held outside banks.

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A customer deposits $200 of currency held outside banks.

Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

# A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

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- Loans and deposits each rise by $120 at origination.

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# A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

# Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.

Source: Federal Reserve Board, H.6 definitions.

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- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.

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# Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.